Product Overview

1.1 Product Name

Storybook Classroom — Interactive Teaching Simulation (绘本互动课堂 — 教学模拟系统)

1.2 Purpose

This is an interactive web-based simulation for the OUM (Open University) course "Teaching English to Young Learners" (Topic 6: Sharing Books with Young Children). Students (pre-service teachers) practice three methods of sharing literature with young children in a simulated kindergarten classroom, receiving immediate rule-based feedback on their choices.

1.3 Target Users

- University students studying early childhood English education
- Pre-service kindergarten English teachers

1.4 Core Experience

The student plays as a kindergarten teacher, conducting a picture book sharing lesson using "The Very Hungry Caterpillar" with a class of 5 virtual children (ages 4–5). The simulation covers three teaching methods from the textbook, progressing through Phase 1 → Phase 2 → Phase 3 → Final Report.

1.5 Technical Constraints

- All scoring is rule-based (no AI dependency)
- Each phase is 3–5 minutes, total experience ~12 minutes
- Single-page web application
- Desktop-first design (1200px+ width), responsive

2. Overall Architecture

Screen Flow

[Welcome Screen] → [Phase 1: Interactive Storybook Reading] → [Phase 2: Shared Reading] → [Phase 3: Storytelling] → [Final Report]

4. Phase 1: Interactive Storybook Reading

5 sequential scenes, each using the Choice Card Pattern (Section 7.1).

4.1 Scene 1-1: Before Reading

- Phase label: “Phase 1 · Interactive Storybook Reading”
- Title: “Before Reading”
- Book Display: 🐛 on green gradient, “The Very Hungry Caterpillar” by Eric Carle
- Instruction: “How will you introduce this book to create interest?”

Choice Cards:

#	Icon	Text	Rating	Score	Feedback
A	🔮	"Look at the cover! What do you think this caterpillar likes to eat?"	BEST ⭐	3	Prediction questions create curiosity and activate prior knowledge.
B	📋	"Today we're going to read about a caterpillar."	OK ⚠	1	Simple statement, doesn't invite thinking.
C	📄	"Open your books to page 1."	POOR ✗	0	No engagement strategy at all.

4.2 Scene 1-2: During Reading — Page 1

- Book: 🍎 on red gradient, “On Monday he ate through one apple. But he was still hungry.”
- Instruction: “What will you say to keep children engaged?”

Choice Cards:

#	Icon	Text	Rating	Score	Feedback
A	🔮	"What do you think he'll eat tomorrow?"	BEST ⭐	3	Prediction keeps children thinking ahead.
B	🔄	"What did he eat?"	OK ⚠	1	Recall only checks memory.
C	💬	"Apples are fruits."	POOR ✗	0	Statement, not a question.

4.3 Scene 1-3: During Reading — Page 2

- Book: 🍐🍐 on green gradient, “On Tuesday he ate through two pears. But he was still hungry.”
- Instruction: “How will you encourage participation?”

Choice Cards:

#	Icon	Text	Rating	Score	Feedback
A	🔢	"Can you count with me? One... two pears!"	BEST ⭐	3	Participation prompts maintain attention.
B	🎨	"What colour are pears?"	OK ⚠	1	Relevant but off-topic from story flow.
C	⏭️	(Skip to next page without interaction)	POOR ✗	0	Missed opportunity for scaffolding.

4.4 Scene 1-4: During Reading — Page 3

- Book: 🍰🍦🧀 on orange gradient, “On Saturday he ate through one piece of chocolate cake, one ice-cream cone, one pickle, one slice of Swiss cheese...”
- Instruction: “The caterpillar has eaten too much! What do you ask?”

Choice Cards:

#	Icon	Text	Rating	Score	Feedback
A	🤢	"Oh no! Do you think he'll get a tummy ache?"	BEST ⭐	3	Prediction based on personal experience.
B	📝	"Can you name all the foods?"	OK ⚠	1	Too demanding for 4–5 year olds.
C	💬	"That's a lot of food."	POOR ✗	0	Statement with no invitation to respond.

4.5 Scene 1-5: After Reading

- Book: 🦋 on purple gradient, “The caterpillar has turned into a beautiful butterfly!” + “— The End —”
- Instruction: “The story is done. What discussion activity will you choose?”

Choice Cards:

#	Icon	Text	Rating	Score	Feedback
A	👫	"What was your favourite food the caterpillar ate? Tell your friend!"	BEST ⭐	3	Peer discussion is age-appropriate.
B	🔁	"Can anyone retell the whole story?"	OK ⚠	1	Too challenging for most 4-year-olds.
C	➡️	"Let's move to the next activity."	POOR ✗	0	Skipping after-reading wastes opportunity.

5. Phase 2: Shared Reading

Different interaction pattern from Phase 1. 3 sequential tasks within a single “Big Book” view.

5.1 Overall Layout

- Phase label: “Phase 2 · Shared Reading”
- Title: “Big Book”
- Subtitle: “Enlarged text, read along together — focus on repetition, rhythm, and print awareness”

5.2 Task 1: Highlight Repetition Words

- Instruction: “Click on words that show a repetition pattern”
- Display sentence: “On Saturday, he ate through one piece of chocolate cake, one ice-cream cone, one pickle, one slice of Swiss cheese...”
- Each word clickable, toggles yellow highlight
- Correct words (repetition patterns): “he” (idx 2), “ate” (idx 3), “through” (idx 4), “one” (idx 5, 10, 13, 15)
- Scoring: ratio >= 70% → 3 pts; >= 40% → 2 pts; else → 1 pt
- Visual feedback: correct=green, missed=red wavy underline, false positive=red
- Auto-advance after 1.8s

Score Category: "shared" (contributes to max 9)

5.3 Task 2: Identify Book Features

- Instruction: “What feature makes this book ideal for Shared Reading?”
- Options: A) Predictable structure (CORRECT), B) Repetition (CORRECT), C) Long complex sentences (INCORRECT)
- A or B → 3 points; C → 0 points
- Auto-advance after 1.5s

Score Category: "shared"

5.4 Task 3: Left-to-Right Directionality

- Pointing animation: 👉 moves left to right (3s loop)
- Question: “What reading skill are you modeling?”
- Options: A) Left-to-right directionality (CORRECT, 3pts), B) Phonics (0pts), C) Vocabulary (0pts)
- Show “Continue →” button after completion

Score Category: "shared"

6. Phase 3: Storytelling

3 sequential scenes.

6.2 Scene 3-1: Vocal Skills

- Speech: “Let me tell you the story of a very... hungry... caterpillar...”
- Vocal Skill Meters: Intonation (purple), Pitch (blue), Volume (orange) — default 20%

Choice Cards:

#	Icon	Text	Rating	Score	Meters
A	🎭	Slow, mysterious whisper that builds to excitement	BEST ⭐	3	95%/90%/85%
B	😐	Steady, even-paced reading voice	POOR ✗	0	20%/20%/30%
C	📢	Very loud and fast throughout	OK ⚠	1	40%/50%/90%

6.3 Scene 3-2: Body Language

- Speech: “The caterpillar ate and ate and ATE!”

Choice Cards:

#	Icon	Text	Rating	Score
A	🙌	Growing hand gestures — small then BIG! Lean forward	BEST ⭐	3
B	🧍	Stand still, hands at sides	POOR ✗	0
C	😮	Facial expressions only	OK ⚠	1

6.4 Scene 3-3: Visual Aids

- Speech: “He ate through one apple, two pears, three plums...”

Choice Cards:

#	Icon	Text	Rating	Score
A	🍎	Real fruit props/flashcards children can touch	BEST ⭐	3
B	💻	PowerPoint slideshow	OK ⚠	1
C	🚫	No visual aids	POOR ✗	0

8. Final Report

8.1 Score Categories

Category	Max	Source
Before Reading	3	Phase 1, Scene 1
During Reading	9	Phase 1, Scenes 2–4
After Reading	3	Phase 1, Scene 5
Shared Reading	9	Phase 2, Tasks 1–3
Storytelling	9	Phase 3, Scenes 1–3
TOTAL	33	

8.2 Rating Calculation

Percentage	Stars	Title
>= 85% (28+)	★★★★★	Master Storyteller🏆
>= 75% (25+)	★★★★☆	Master Storyteller 🏆
>= 60% (20+)	★★★☆☆	Reading Guide 🌟
>= 40% (14+)	★★☆☆☆	Story Explorer📖
< 40%	★☆☆☆☆	Story Explorer 📖